

# Real-time video anomaly detection — 11 classes, open-set

One frozen encoder, encoded once, read three ways. No backbone fine-tuning.

57.7

benchmark score

18–58×

realtime

125 s

all 28 videos

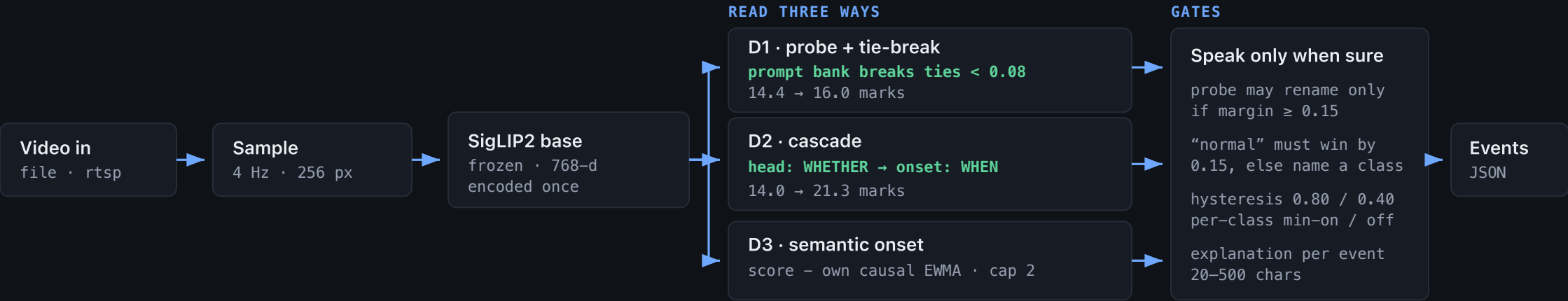
2.4 MB

trained weights

A2000

12 GB GPU

PIPELINE



WHY THIS SHAPE

- **Frozen encoder, one pass.** Features computed once, read three ways — why it beats realtime on a modest GPU.
- **Prompts are data, not weights.** A twelfth class is a sentence, not a retrain.
- **Score against normal.** Margin over the best of ~70 normal prompts; raw similarity just describes the scene.
- **Long video needs a baseline.** D3 keys on the rise over a causal 20 s EWMA, so a busy junction isn't permanently anomalous.
- **Locating ≠ naming.** Detector says when, probe says what — 63% vs 33% on class.

ENCODERS EVALUATED — IDENTICAL SCORING CODE

| BACKBONE                 | DIM | D1   |       |
|--------------------------|-----|------|-------|
| SigLIP2-base/224 shipped |     | 768  | 45.8% |
| SigLIP2-so400m/384       |     | 1152 | 54.2% |

# Nine full runs, every one scored on the live benchmark

Each row is one complete pipeline run over all 28 videos. Nothing estimated, nothing assembled from parts.

## EVERY SUBMISSION, AND WHAT IT SCORED

| #                                                                                         | WHAT WE CHANGED                                       | D1 /25 | D2 /35 | D3 /40 | MARKS |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------|--------|--------|--------|-------|
| 1                                                                                         | Baseline — probe · temporal head · onset              | 11.6   | 14.0   | 16.9   | 42.5  |
| 2                                                                                         | "normal" must win by a margin, else name the anomaly  | 14.4   | 14.0   | 16.9   | 45.3  |
| 3                                                                                         | Half-max boundary refinement                          | 14.4   | 14.0   | 8.0    | 36.4  |
| 4                                                                                         | Onset instead of the head on D2                       | 14.4   | 12.6   | 16.9   | 43.9  |
| 5                                                                                         | Adaptive per-video D2 thresholds                      | 14.4   | 5.3    | 16.9   | 36.6  |
| 6                                                                                         | Tighter D2 hysteresis                                 | 14.4   | 14.0   | 16.9   | 45.3  |
| 7                                                                                         | D2 cascade — head decides whether, onset decides when | 14.4   | 21.3   | 16.9   | 52.6  |
| 8                                                                                         | Keep 2 intervals per long video, not 3                | 14.4   | 21.3   | 17.9   | 53.6  |
| 9                                                                                         | Prompt bank breaks the probe's ties                   | 16.0   | 21.3   | 17.9   | 55.3  |
| ↓ organisers introduced false-positive penalties — everything below is scored differently |                                                       |        |        |        |       |
| 10                                                                                        | Run 9 re-scored, unchanged, under the new rules       | 16.0   | 20.5   | 14.4   | 50.9  |
| 11                                                                                        | D3 intervals ranked by the head, not by duration      | 16.0   | 20.5   | 15.2   | 51.7  |
| 12                                                                                        | One D3 interval per video — false alarms 6 → 2        | 16.0   | 20.5   | 17.7   | 54.2  |
| 13                                                                                        | Also cut D2 from 9 events to 6                        | 16.0   | 20.0   | 17.7   | 53.7  |

Plus a reasoning bonus of +3.5/5 throughout — **57.7 total**. The benchmark changed mid-event: run 9 scored 55.3 before the penalty and 50.9 after, unchanged. Any delta measured across that line is meaningless.

## WHAT WORKED

- **Run 7, the biggest single gain.** The head matched 0 of 12 events but correctly stayed silent on the one normal video; the onset localised well but fired on it. Chaining them collected both — **14.0 → 21.3**.
- **Runs 2 and 9, both on D1.** One clip read normal 0.132 vs smoke 0.131. Letting a second component break ties like that took D1 **46.4% → 64.1%**.

## WHAT DIDN'T

- **Run 3 halved D3.** We tightened intervals to their peak width; the real events run 38–125 s, so tidy spans miss them under tIoU 0.5.
- **Run 5 cost a whole video.** Forcing every clip to fire lost 8.7 marks — almost exactly the 8.75 of the one normal video.